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22201942

Section: 28

Semest: Fall-2023

Course: Mat120

① Ans. to the qno-01

22201942

$$y'' - 6y' + 25y = 50t^3 - 36t^2 - 63t + 18$$

Homogeneous Part:-

$$m^2 e^{mx} - 6m e^{mx} + 25e^{mx} = 0$$

$$m^2 - 6m + 25 = 0$$

$$m = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(1)(25)}}{2}$$

$$= \frac{6 \pm \sqrt{36 - 100}}{2}$$

$$= \frac{6 \pm \sqrt{64}}{2}$$

$$= 3 \pm 4i$$

$$y = e^{\alpha x} \quad \alpha = 3 \quad \beta = 4$$

$$y = e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$$

$$y = e^{3x} (c_1 \cos 4x + c_2 \sin 4x)$$

Particular:-

$$y_p = 50t^3 - 36t^2 - 63t + 18$$

$$Ax^3 + Bx^2 + Cx + D =$$

$$50t^3 - 36t^2 - 63t + 18$$

$$y_p =$$

Ans: $y_p = A + 3t + Ct^2 + Dt^3$

$$(1) y_p = A + 3t + Ct^2 + Dt^3$$

$$y_p' = B + 2Ct + 3Dt^2$$

$$y_p'' = 2C + 6Dt$$

$$2C + 6Dt = -6(B + 2Ct + 3Dt^2) + 25(A + Bt + Ct^2 + Dt^3)$$

$$\Rightarrow 50t^3 - 36t^2 - 63t + 18$$

$$(250)t^3 + (25C - 18D)t^2 + (25B - 12C)t + (25A - 6B + 2C) = 50t^3 - 36t^2 - 63t + 18$$

25D = 50

$$25D = 50$$

$$25C - 18D = -36$$

$$25B - 12C + 6D = -63$$

$$25A - 6B + 2C = 18$$

$$D = \frac{50}{25} = 2$$

$$25C - 18 \times 2 = -36$$

$$\boxed{C = 0}$$

$$25B - 12 \times 0 + 6 \times 2 = -63$$

$$25B = -63 - 12$$

$$B = \frac{-75}{25} = -3$$

$$25(A) - 6 \times (-3) + 2 \times (0) = 18$$

$$25A = 0$$

$$A = 0$$

$$y_p = 2t^3 - 3t$$

$$y_{total} = \cancel{2t^3 - 3t} e^{3t} (c_1 \sin 4t + c_2 \cos 4t) + 2t^3 - 3t$$

⑤ Ans-2

(a) $3x + 2y - 2 \geq 0$
 $y \geq x^2 \quad x \geq y^2$

$$3x + 2y - 2 \geq 0$$

$$2 \geq 3x + 2y$$

$$y \geq x^2 \quad x \geq y^2$$

$$x \geq (x^2)^2$$

$$x^4 - x \geq 0$$

$$x(x^3 - 1) \geq 0$$

$$x = 0, 1$$

x

$$0 \leq x \leq 1 \quad , \quad x^2 \leq y \leq \sqrt{x}$$

$$V = \int_0^1 \int_{x^2}^{\sqrt{x}} dy dx$$

$$\int_0^1 \int_{x^2}^{\sqrt{x}} (3x + 2y) dy dx$$

$$\int_0^1 \left[3xy \right]_{x^2}^{\sqrt{x}} + \left[\frac{2y^2}{2} \right]_{x^2}^{\sqrt{x}} dx$$

$$= \int_0^1 3x^3 - 3x\sqrt{x} + (x - x^4) dx$$

$$= \left[\frac{3x^4}{4} - \frac{3x^{\frac{3}{2}+1}}{\frac{3+2}{2}} + \left[\frac{x^{\frac{2}{2}+1}}{2} - \frac{x^{\frac{5}{2}+1}}{\frac{5}{2}+1} \right]_0^1 \right]$$

$$⑥ = \frac{2}{5} \left[\frac{3 \times \frac{2}{2}}{\frac{3}{4}} + \frac{1}{2} - \frac{3 \times 1}{4} - \frac{1}{5} \right] \quad 22201942$$

$$2 \left[\frac{6}{5} + \frac{1}{2} - \frac{3(1)}{4} - \frac{1}{5} \right]$$

$$2 \frac{24 + 10 - 15 - 4}{20}$$

$$= \frac{3}{4}$$

Ans. to q no - 03

T

T $\rightarrow$ temp of the body

$$T_x = 100$$

we have,

$$\frac{dT}{dt} = -k(T - T_x)$$

$$\frac{dT}{dt} + kT = kT_x$$

where k is constant the constant is proportional.

$$\therefore \frac{dT}{dt} + kT = 100k$$

$$T = ce^{-kt} + 100 \quad \text{--- (1)}$$

Here $T = 50$ at $t = 0$

$$50 = ce^{-k(0)} + 100$$

$$\text{So that } c = -50$$

Now put this

$$T = -50e^{-kt} + 100 \quad \text{--- (11)}$$

If $t = 6$ $T = 60$

$$60 = -50e^{-5k} + 100$$

$$-40 = -50e^{-5k}$$

$$k = \frac{1}{5} \ln \frac{40}{50} = \ln e^{-5k}$$

$$\frac{1}{5k} = \ln \frac{4}{5}$$

$$k = \frac{1}{5} \ln \left(\frac{4}{5} \right)$$

$$T = -50e^{\frac{1}{5} \ln \left(\frac{4}{5} \right) t} + 100 \quad \text{--- (11)}$$

Q

$$If = T = 75$$

$$\frac{1}{2} = e^{\frac{1}{5} \ln(1/5)t}$$

$$\ln\left(\frac{1}{2}\right) = \frac{1}{5} \ln(1/5)t$$

$$5 \ln(1/2) = \ln(1/5)t$$

$$t = 15.53 \text{ min}$$

(c)

T when $t = 20$

$$T = -50 e^{(1/5) \ln(1/5) \times 20} + 100$$

$$T = -50 e^{(1/5) \ln(1/5) \times 20} + 100$$

$$T = -50 \left(\frac{1/5}{1/2} \right)^2 + 100$$

$$T = -50 \left(\frac{1/5}{1/2} \right)^2 + 100$$

$$79.52$$

Ans. to the qno-4

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_{2-\sqrt{4-x^2-y^2}}^{2+\sqrt{4-x^2-y^2}} (x^2+y^2+z^2)^{3/2} dz dy dx$$

$$\int_0^\pi \int_0^{2\sqrt{2}} \int_0^{2\sqrt{2}} \rho^{3/2} \rho^2 \sin\phi d\rho d\phi d\theta$$



$$z = 2 + \sqrt{4-x^2-y^2}$$

$$z^2 = 4 + 4 - x^2 - y^2$$

$$\rho^2 = 8$$

$$\text{full circle } \rho = 2\sqrt{2}$$

$$\theta = 2\pi$$

$$\phi = 0$$

$$\theta = 0$$

From z limit this is hemisphere

$$\phi = \frac{\pi}{2} + \frac{\pi}{2}$$

$$\phi = \pi$$

$$\phi = 0$$

Now

$$\int_0^{2\pi} \int_0^\pi \int_0^{2\sqrt{2}} \rho^{3/2} \rho^2 \sin\phi d\rho d\phi d\theta$$

$$\int_0^{2\pi} \int_0^\pi \int_0^{2\sqrt{2}} \rho^3 \sin\phi d\rho d\phi d\theta$$

$$\int_0^{2\pi} \int_0^{\pi} \left[\frac{8^4}{4} \right]^{2\pi} \sin \phi d\phi d\theta$$

$$= -16 \int_0^{2\pi} [\cos \phi]_0^{\pi} d\theta$$

$$= -16 \left[(\cos \pi - \cos 0) \times 2\pi \right]$$

$$= -16 (-2 \times 2\pi)$$

$$= 64\pi$$

Q1) Ans to the q no-03

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$$\frac{dy}{dx} = \frac{x^2 + y^2 - 5}{y + xy}$$

$$(y + xy) dy = dx (x^2 + y^2 - 5)$$

$$(y + xy) dy - (x^2 + y^2 - 5) dx$$

$$M = y + xy$$

$$\frac{dM}{dx} = 1 + y$$

$$N = x^2 + y^2 - 5$$

$$\frac{dN}{dy} = 0 + 2y$$

$$\frac{dM}{dx} \neq \frac{dN}{dy} \text{ (Not exact)}$$

$$\frac{Mx - Ny}{N} = \frac{1 - 2y}{x^2 + y^2 - 5}$$

$$\frac{Nx - My}{M} = \frac{2y - 1}{y + xy} = \frac{2y - 1}{y(1+x)}$$

Both the quotient depends on both variable x and y .

~~This is not a linear equation. The~~
~~depend variable~~

$$y_p = 2t^3 - 3t$$

$$y_{total} = \cancel{2t^3 - 3t} e^{2t} (c_1 \sin 4t + c_2 \cos 4t) + 2t^3 - 3t$$

3) 2nd order eqn

$$I f = p(x) = e^{\int \frac{1}{1+x} dx}$$

$$= (1+x)$$

$$(1+x)(y+xy) dy - x(x^2+y^2-5)(x+y) = 0$$

$$\therefore dy(xy + xy^2 + y^2 + y) - dx(x^2 + xy^2 - 5x + x^2 + y^2 - 5) = 0$$

$$\frac{dy}{dx} (2xy + x^2 + y^2 + 5) = dx(x^2 + xy^2 - 5x + x^2 + y^2 - 5)$$

$$df(x,y) = \int x^3 + x^2 y^2 + xy^2 - \sin - 5) dx$$

$$= \frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2 y^2}{2} - \frac{\sin x}{2} + C(y)$$

$$\frac{df}{dy} = 2xy + x^2 y + y =$$

$$2xy + x^2 y + y = x^2 y + y'(y)$$

$$g'(y) = 2xy + x^2 y$$

$$g(y) = \int (2xy) dy = \frac{xy^2}{2} + \frac{y^3}{3} + C(x)$$

$$f(x,y) = \frac{x^4}{4} + \frac{x^3}{3} + \frac{x^2 y^2}{2} - \frac{\sin x}{2} + xy^2 + \frac{y^3}{3}$$

⑤ ⑥ $y''' - 2x^4 y' + x^5 y = 50t^3 \sin t$ 22201942

$\frac{d^3 y}{dx^3} - 2x^4 \frac{dy}{dx} + x^5 y = 50t^3 \sin t$

① linear equation the dependent variable's power is $\neq 1$.

(ii) this is not a homogeneous equation

(iii) this 3rd order differential equation.

(iv) degree 1: